

ACP-Gini: Path-dependent Correlation Penalization for Decision Tree Induction

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Abstract

Decision trees avoid coefficient estimation under multicollinearity, yet greedy induction still splits on interchangeable members of correlated predictor groups, which yields redundant feature sets and disperses importance over proxies. We introduce ACP-Gini, a single-tree splitting criterion that multiplies each candidate's raw Gini gain by a product of correlation penalties against the distinct features already used on the root-to-node path. It reduces exactly to CART at $\alpha = 0$ and keeps raw gain for stopping. On eight binary-classification data sets evaluated over 15 held-out folds, ACP-Gini lowered bootstrap importance-weighted redundancy in every data set (median reduction 11.5%, range 0.9% to 29.6%; all Holm-adjusted $p \leq 0.0013$), and the reduction persisted under Spearman and mutual-information measures that the penalty does not use. No accuracy change was significant, but the 95% intervals do not exclude losses of several points on the smallest data sets (up to 4.9 points for Sonar, 8.3 with fully nested tuning of α). The benefit has costs. Used-feature stability fell on all eight data sets, and importance-rank stability fell significantly on three while rising on Wine. A correlation-aware penalty over the whole tree gave the same average redundancy reduction as the path-local penalty, and tuned VIF or cluster filters matched it while discarding features. ACP-Gini is therefore a controllable accuracy-redundancy-stability trade-off that keeps every feature available to every branch, not a uniform improvement. Code is available at <https://github.com/AI-Peak/acp-gini>.

Keywords: Decision trees, multicollinearity, feature redundancy, model stability, interpretable machine learning

1 Introduction

Decision trees remain a default choice for interpretable machine learning because they avoid the simultaneous coefficient estimation that multicollinearity destabilizes in linear models. That immunity is narrower than it appears. A greedy tree must still choose split features from blocks of correlated predictors whose impurity gains are nearly interchangeable. Small sample changes can then switch the chosen representative, and

a single tree can split on different members of one correlated block along different branches. The result is an internally redundant feature set and an importance profile spread across proxies, which hinders audit. Correlated predictors are known to make importance rankings unreliable [10, 25], but most remedies operate outside the tree.

Existing remedies fall into four groups. Bagging [1] and random forests [2] stabilize predictions but give up the single, directly readable model, and conditional or randomized importance measures [9, 15, 18, 24] diagnose ensembles after training. Filter methods remove redundant variables before learning, either globally by variance-inflation pruning [7] or by relevance-redundancy criteria such as CFS, FCBF and mRMR [11, 20, 26]; both discard a feature for every branch even when it carries conditional information on one. Regularized trees [5, 6] penalize the use of new features globally and do not measure correlation. Finally, oblique and optimal sparse trees [13, 16] change the model class at higher computational cost. To our knowledge none of these injects the pairwise correlation between a candidate and the features already used above it into the split score of an ordinary axis-aligned tree, and none keeps every feature available to every branch while doing so.

We propose ACP-Gini (Ancestor-Correlation Penalized Gini), which discounts a candidate’s Gini gain by its correlation with the distinct features already used on the current root-to-node path. The criterion equals CART at $\alpha = 0$, exempts a feature from penalizing itself so that continuous variables can split repeatedly, and uses raw gain for stopping so that the penalty only reorders candidates. Table 1 positions it against the closest single-tree alternatives. Whether the path-local scope matters, as opposed to the correlation-aware penalty itself, is an empirical question that Section 3.7 tests.

Table 1 Positioning of the evaluated single-tree approaches.

Method	Single tree	Corr.-aware	Path-local	No preproc.	All features usable
VIF+CART	Yes	Yes	No	No	No
Cluster+CART	Yes	Yes	No	No	No
RRF-style	Yes	No	No	Yes	Yes
ACP-Gini	Yes	Yes	Yes	Yes	Yes

We ask four questions. Does the penalty reduce redundancy in the selected feature set and along individual decision paths? What does it cost in accuracy? How does it affect the bootstrap stability of selected features and importance rankings? And do the answers hold across data sets, depth caps and correlation estimators? Our contributions are as follows.

- We define the path-local ancestor-correlation penalty with no-self-penalty and raw-gain stopping rules, and prove its monotonicity, boundedness, informative-split preservation and duplicate-elimination properties.
- We add path-level redundancy and group-level stability diagnostics to the selected-set redundancy and bootstrap stability metrics, and check redundancy with Spearman and mutual-information measures that the penalty does not use.
- We benchmark ACP-Gini against CART, VIF+CART, a supervised Cluster+CART filter and an RRF-style tree on a synthetic generator and eight real data sets, with corrected resampled inference, Holm adjustment, fully nested tuning, depth and

estimator ablations, a tree-global scope ablation and threshold sweeps for both filters. The evidence supports a consistent redundancy reduction without a significant accuracy change, a real stability cost, and a conclusion that the penalty, rather than its path-local scope, accounts for the effect on these data.

The implementation, baselines and replication scripts are at <https://github.com/AI-Peak/acp-gini>. We position the work within the case for auditable models in consequential domains [21].

2 Materials and Methods

2.1 Data sets

We use eight numeric binary-classification data sets and a synthetic generator. WDBC [19, 23], Wine Quality [4], Ionosphere, Sonar and Pima [22] were used in the original design; Wine Quality is binarized at quality ≥ 6 (63.3% positive). We added three scikit-learn data sets, none of which required download. Diabetes [8] is a classical multicollinear design, binarized at the median response. Cultivar is the UCI wine-cultivar data [14] with cultivar 0 against the rest. Digits38 contains the confusable digit pair 3 versus 8 from the UCI optical-digits data [14], with constant pixels removed. Table 2 lists sizes and the share of feature pairs with $|r| > 0.7$. Only WDBC is strongly collinear by this measure (16.1%); Diabetes, Sonar and Cultivar are mildly collinear, and Wine, Digits38, Ionosphere and Pima serve as low-collinearity controls.

The synthetic generator draws five independent Gaussian signals with weights 1.0, 0.8, 0.6, 0.4, 0.2 and three noisy copies of each, $\rho Z + \sqrt{1 - \rho^2} \epsilon$, with $\rho \in \{0.7, 0.8, 0.9, 0.95, 0.99\}$, plus five pure-noise features. The generator has 1000 samples per repeat, a 70/30 stratified split, label noise with standard deviation 0.5, and each setting is repeated ten times. Missing values in the UCI sets are filled with the column median and Pima is used as distributed, with zeros kept as values.

Table 2 Data-set statistics and collinearity diagnostics.

Data set	Samples	Features	Positive rate	Pairs with $ r > 0.7$
WDBC	569	30	0.627	0.161
Wine	6497	11	0.633	0.018
Ionosphere	351	34	0.359	0.005
Sonar	208	60	0.534	0.031
Pima	768	8	0.349	0.000
Diabetes	442	10	0.500	0.044
Cultivar	178	13	0.331	0.026
Digits38	357	54	0.487	0.007

2.2 ACP-Gini

For a node T with n_T samples and a candidate split into T_L and T_R , the Gini impurity and raw gain are

$$G(T) = 1 - \sum_k p_k(T)^2, \quad (1)$$

$$\Delta G(X_j, t) = G(T) - \frac{n_L}{n_T} G(T_L) - \frac{n_R}{n_T} G(T_R). \quad (2)$$

Let $A(T)$ be the distinct feature indices used on the path from the root to T . The candidate is not penalized against itself, so

$$A_j(T) = \{p \in A(T) : p \neq j\}, \quad (3)$$

$$P(X_j, T) = \prod_{p \in A_j(T)} (1 - \alpha |r(X_j, X_p)|), \quad (4)$$

$$S(X_j, t) = \Delta G(X_j, t) P(X_j, T). \quad (5)$$

The selected split maximizes $(S, \Delta G, -j)$ lexicographically, so ties go to larger raw gain and then to the lower feature index. Candidate thresholds are midpoints between distinct sorted values, evaluated from cumulative class counts. Induction stops at a node only if the selected raw gain is at most 10^{-15} .

The correlation matrix is Pearson by default and computed once on the training fold; Spearman ranks are supported. Undefined constant-column correlations are set to zero. A node-local scope recomputes correlations on the samples at the node when it holds at least 30, and otherwise falls back to the global matrix.

Selecting α .

We evaluate $\alpha \in \{0.2, 0.4, 0.6, 0.8, 1.0\}$ by stratified inner cross-validation. We keep the values whose inner macro-F1 lies within one standard deviation of the CART mean and, among them, choose the value with the highest bootstrap importance-rank correlation. The data-set value is the lower quartile of the fold-level choices rounded down to the grid, a conservative consensus. For the RRF-style baseline $\lambda \in \{0.5, 0.7, 0.9\}$ is tuned the same way with the upper quartile. Inner cross-validation uses three folds and ten bootstrap trees per candidate, ties go to the smaller α , and the F1 tolerance is the standard deviation of the three inner-fold CART scores. The consensus value is chosen using all training folds of a data set, so we also run a fully nested variant in which α is chosen in each outer fold using that fold's training data only [3].

2.3 Theoretical properties

Write $P_A(X_j)$ for the penalty when the ancestor set is A .

Property 1 (monotonicity in the ancestor set). For $A \subseteq A'$, any candidate X_j and $\alpha \in [0, 1]$, $P_{A'}(X_j) \leq P_A(X_j)$ and hence $S_{A'}(X_j, t) \leq S_A(X_j, t)$.

Proof. $A \subseteq A'$ gives $A_j \subseteq A'_j$, so $P_{A'}(X_j) = P_A(X_j) \prod_{p \in A'_j \setminus A_j} (1 - \alpha |r(X_j, X_p)|)$. Each factor lies in $[0, 1]$ because $\alpha \in [0, 1]$ and $|r| \in [0, 1]$, so $P_{A'}(X_j) \leq P_A(X_j)$. Multiplying by $\Delta G(X_j, t) \geq 0$ preserves the inequality. $\square$

A candidate's score therefore never rises as the path deepens. An arithmetic-mean aggregation lacks this property, because appending a weakly correlated ancestor raises the mean of the factors and weakens the penalty.

Property 2 (monotonicity in correlation). For fixed other ancestors, $P(X_j, T)$ is non-increasing in $|r(X_j, X_p)|$ for every $p \in A_j(T)$.

Proof. The factor $1 - \alpha |r(X_j, X_p)|$ has derivative $-\alpha \leq 0$ in $|r|$, and the remaining factors are non-negative. $\square$

Property 3 (boundedness). $0 \leq P(X_j, T) \leq 1$ for all X_j , $A(T)$ and $\alpha \in [0, 1]$, and $P(X_j, T) = 1$ when $\alpha = 0$ or $A_j(T) = \emptyset$.

Proof. Each factor lies in $[0, 1]$, so does their product, and the empty product is 1. $\square$

Lemma 1 (informative splits are preserved). If some candidate at T has $\Delta G > 0$, the candidate ACP-Gini selects has $\Delta G > 0$, for every $\alpha \in [0, 1]$.

Proof. Let c be the candidate that maximizes $(S, \Delta G, -j)$ lexicographically. If some candidate has $S > 0$, then $S(c) > 0$, and $S = \Delta G P$ with $P \leq 1$ gives $\Delta G(c) > 0$. If every candidate has $S = 0$, which requires $\alpha = 1$ and some $|r| = 1$ for each informative candidate, the first key is tied and the second key selects the candidate of maximal raw gain, which is positive by assumption. $\square$

Induction stops only when the selected raw gain is at most the numerical tolerance 10^{-15} , so the penalty reorders candidates but does not turn a node with a positive-gain split into a leaf, up to that tolerance.

Property 4 (exact duplicates of ancestors). Let $j \in A(T)$ and let X_k be a feature with $|r(X_k, X_j)| = 1$ and $k \notin A(T)$. At $\alpha = 1$, $P(X_k, T) = 0$, so X_k is selected at T only if every candidate with positive raw gain is also fully penalized. For $\alpha < 1$, $P(X_k, T) \leq 1 - \alpha$.

Proof. The factor for $p = j \in A_k(T)$ equals $1 - \alpha \cdot 1$ and all other factors are at most 1, which gives both bounds. At $\alpha = 1$, $S(X_k, t) = 0$, so X_k can be selected only if no candidate has $S > 0$, and then the second key (raw gain) decides. $\square$

Property 4 explains the behaviour on exact copies; for correlations below one the penalty is graded, and the paper makes no stability guarantee.

Complexity.

CART costs $O(nd \log n)$ per tree level. For each node ACP-Gini looks up the correlations of d candidates with $|A(T)| \leq h$ ancestors, an extra $O(dh)$ per node, and computes the global correlation matrix once in $O(nd^2)$. With $h \leq 10$ in our experiments the lookup is negligible against the split scan.

2.4 Baselines and evaluation protocol

All trees use maximum depth 6 and minimum leaf size 5 unless a depth ablation states otherwise. We compare ACP-Gini with four trees.

- **CART:** the same code at $\alpha = 0$; held-out accuracy differs from scikit-learn’s classifier by at most 0.010 over 15 sanity checks.
- **VIF+CART:** features are removed iteratively while the largest variance inflation factor is at least 10, then a CART tree is grown on the rest.
- **Cluster+CART:** features are clustered with complete linkage so that every within-cluster pair has $|r| \geq 0.7$, the feature with the highest univariate Gini gain is kept per cluster inside the training fold, and CART is grown on the representatives.
- **RRF-style:** candidates not yet used anywhere in the tree have their gain multiplied by λ [5].

A 100-tree random forest with depth 6 and leaf size 5 [2] is reported as a non-interpretable accuracy reference only. The VIF and cluster thresholds are fixed at conventional values and not tuned, whereas α is tuned; Section 4.1 returns to this asymmetry.

Evaluation uses three shuffled stratified five-fold cross-validation repeats (seeds 42, 43, 44), giving 15 paired held-out folds per data set, and 50 bootstrap refits of the

training portion of each fold for stability and redundancy. Unless a caption says otherwise, redundancy numbers are bootstrap averages; the α sweeps, depth and estimator ablations, scope ablation and path-mass bands use one tree fitted on each training fold, so their relative changes are not equal to the bootstrap values (for WDBC at $\alpha = 0.2$, -9.8% against -12.1%).

Metrics.

Predictive performance is accuracy, macro-F1 (the unweighted mean of per-class F1) and AUC. Stability is the Jaccard similarity of used-feature sets and of top-five features, the Spearman correlation of importance vectors (rank correlation) and the split-composition distance, one minus the multiset Jaccard similarity of (depth, feature) pairs. For the used feature set U , the importance-weighted redundancy is

$$R_w = \frac{\sum_{i < j \in U} I_i I_j |r_{ij}|}{\sum_{i < j \in U} I_i I_j}, \quad (6)$$

where I_i is the normalized impurity-decrease importance (and $R_w = 0$ when a tree uses fewer than two features), and the unweighted set redundancy is the mean $|r_{ij}|$ over pairs in U . Importance vectors include zeros for unused or dropped features, so the rank correlation depends partly on how many features a method leaves unused and should be compared across methods with that in mind. Correlations come from the full data as a fixed post-hoc scale. To avoid judging redundancy only with the Pearson matrix that the penalty uses, we also recompute R_w with Spearman $|\rho|$ and with normalized mutual information on ten quantile bins. Because the penalty acts along paths, we also report the path correlation R_{path} , the leaf-sample-weighted mean of the pairwise $|r|$ among the distinct features on a root-to-leaf path (paths with fewer than two features contribute zero), and the share of leaf samples whose path contains a pair with $|r| > 0.7$. For group-level stability, features are partitioned by complete-linkage clustering on $1 - |r|$ with the cut at 0.3, so every within-group pair has $|r| \geq 0.7$. Group importance is the summed importance of a group, and the group-level rank correlation and group Jaccard are computed over bootstrap trees like their feature-level counterparts. On the synthetic data, group coverage is the fraction of signal groups among the top five features by importance, group concentration is the mean over groups of $\max_{j \in g} I_j / \sum_{j \in g} I_j$, and noise importance is the importance summed over the noise features.

Inference.

The 15 folds overlap in their training data, so a plain paired t -test or Wilcoxon test overstates significance. Our primary tests use the corrected resampled t -test of Nadeau and Bengio [17], whose variance factor is $1/J + n_{\text{test}}/n_{\text{train}} = 1/15 + 1/4$, with 14 degrees of freedom and 95% intervals on the mean paired difference. The correction was derived for differences in test error; we apply it unchanged to the bootstrap-derived redundancy and stability metrics as a conservative approximation, which is a limitation (Section 4.1). p -values are adjusted across the eight data sets within each metric by the Holm procedure [12]. As a data-set-level check we count the data sets in which a metric is lower under ACP-Gini and apply a two-sided sign test; ties count as not lower. Paired Wilcoxon tests are retained in the repository as a secondary check.

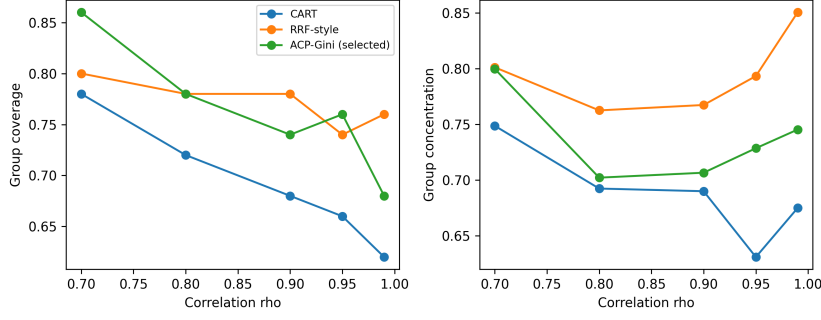


Fig. 1 Synthetic group coverage and concentration (mean of ten repeats). RRF-style attains high concentration by feature reuse; Fig. 3 and Table 8 test transfer to real data.

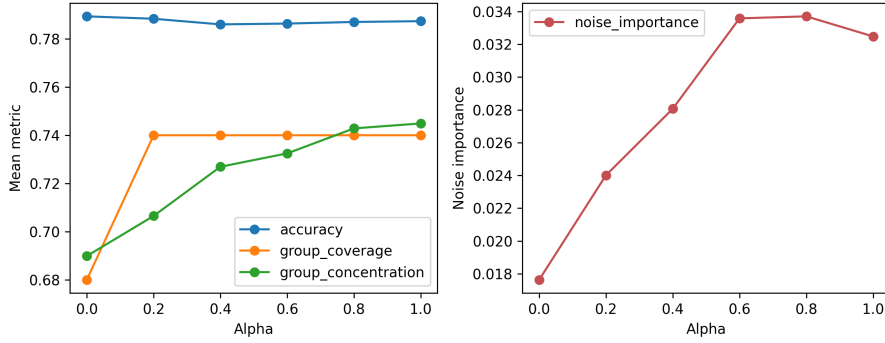


Fig. 2 ACP-Gini sensitivity at $\rho = 0.9$: accuracy and group metrics (left), and the noise-importance side effect (right).

Because five of the eight data sets are low-collinearity controls, we do not pool across data sets into one effect.

3 Results

3.1 Synthetic recovery of correlated groups

At $\rho = 0.9$, increasing α from 0 to 1 raised group coverage from 0.68 to 0.74 and group concentration from 0.690 to 0.745 (Fig. 1). Across the five correlation levels, coverage rose by 0.04 to 0.10 and concentration by 0.026 to 0.098. Pooling the 50 paired settings at $\alpha = 1$, the mean gain over CART was 0.060 in coverage (95% interval 0.031 to 0.089, improved in 32% of runs because coverage moves in steps of 0.2) and 0.061 in concentration (0.041 to 0.082, improved in 86% of runs). Accuracy was unchanged (mean -0.000 , 0.007 below to 0.006 above), and noise importance rose at every ρ , by 0.008 to 0.028 (from 0.018 to 0.032 at $\rho = 0.9$, Fig. 2). These synthetic runs fix $\alpha = 1$ and do not use the tuning protocol applied to real data. The RRF-style baseline reached higher coverage than ACP-Gini at $\rho \leq 0.9$ (0.80, 0.78 and 0.78 against 0.84, 0.76 and 0.74 for ACP-Gini at $\rho = 0.7, 0.8$ and 0.9 , so it is ahead only from $\rho = 0.8$) and higher concentration at $\rho \geq 0.9$ (0.767 against 0.745 at $\rho = 0.9$, 0.851 against 0.743 at $\rho = 0.99$) by rewarding reuse of already-selected features. The next sections test whether that advantage transfers to real data.

Table 3 Accuracy change of ACP-Gini relative to CART (percentage points) with 95% corrected-resampled intervals, for the consensus α and for α tuned inside each outer fold.

Data set	Consensus α	Nested α
WDBC	+0.06 [-1.66, +1.78]	-0.06 [-2.48, +2.37]
Diabetes	+0.23 [-3.17, +3.63]	-0.00 [-4.42, +4.42]
Sonar	-0.65 [-4.85, +3.55]	-2.24 [-8.32, +3.84]
Cultivar	-0.37 [-2.52, +1.79]	-0.18 [-2.55, +2.19]
Wine	-0.10 [-1.39, +1.19]	-0.05 [-1.31, +1.21]
Digits38	+0.37 [-2.30, +3.05]	+0.47 [-2.22, +3.15]
Ionosphere	+0.09 [-2.77, +2.96]	-0.38 [-3.39, +2.63]
Pima	-0.43 [-1.49, +0.62]	+0.00 [-3.30, +3.30]

Table 4 Selected data-set-level penalty parameters.

Data set	α	λ
Cultivar	0.200	0.700
Diabetes	0.400	0.900
Digits38	0.200	0.900
Ionosphere	0.400	0.900
Pima	0.200	0.900
Sonar	0.200	0.900
WDBC	0.200	0.700
Wine	1.000	0.900

3.2 Predictive performance

Selected parameters are in Table 4. Across the eight data sets the mean accuracy was 0.824 for ACP-Gini and 0.825 for CART (Table 5), and none of the eight corrected tests was significant (Table 7). A non-significant test is not evidence of equivalence, so Table 3 gives the intervals. For the consensus α the lower 95% limit of the accuracy change ranged from -1.4 to -4.9 percentage points, which supports non-inferiority at a 5-point margin on every data set but at a 2-point margin only on WDBC, Wine and Pima. The consensus α is chosen with all training folds of a data set, so the nested variant, which tunes α inside each outer fold, is the less biased estimate. Its mean accuracy was 0.822 and its lower limits were wider, from -1.3 to -4.4 points on seven data sets and -8.3 points on Sonar (nested accuracy 0.715 against 0.737 for CART). Losses of a few points on small data sets are therefore not excluded, and the claim we can support is the absence of a detectable change, not equivalence. The random-forest reference reached 0.865, so the cost of staying with a single tree, which ACP-Gini shares with CART, is about four points on average and up to 7.5 points on Sonar (Table 6).

3.3 Selected-set and path-level redundancy

ACP-Gini lowered bootstrap importance-weighted redundancy in all eight data sets (sign test $p = 0.008$; Fig. 3a, Table 7). The relative reduction had a median of 11.5% and ranged from 0.9% on Pima to 29.6% on Wine, every Holm-adjusted p was at most 0.0013, and the paired standardized effects ranged from -2.2 to -18.0 . The same held at the level of single paths. Mean path correlation fell in all eight data sets (Holm

Table 5 Mean over the eight data sets of the per-data-set fold means. Bold marks the best interpretable method per column (random forest excluded); redundancy and path columns are lower-is-better. Group corr. is the group-level importance rank correlation, R_w the bootstrap importance-weighted redundancy and Path $|r|$ the bootstrap path correlation.

Method	Acc.	F1	AUC	R_w	Path $ r $	Rank corr.	Group corr.
CART	0.825	0.814	0.857	0.319	0.286	0.454	0.535
VIF+CART	0.823	0.812	0.860	0.277	0.255	0.488	0.502
Cluster+CART	0.822	0.812	0.858	0.303	0.271	0.461	0.500
RRF-style	0.825	0.814	0.858	0.316	0.278	0.454	0.540
ACP-Gini	0.824	0.813	0.857	0.284	0.243	0.453	0.498
ACP-Gini (nested)	0.822	0.811	0.857	0.268	0.225	0.450	0.484
Random forest (ref.)	0.865	0.855	0.919	—	—	—	—

Table 6 Held-out accuracy by method and data set (15 folds).

Data set	CART	VIF+CART	Cluster+CART	RRF-style	ACP-Gini	ACP-Gini (nested)	Random forest (ref.)
WDBC	0.934	0.931	0.931	0.941	0.934	0.933	0.954
Diabetes	0.674	0.686	0.667	0.683	0.676	0.674	0.731
Sonar	0.737	0.715	0.723	0.726	0.731	0.715	0.806
Cultivar	0.951	0.951	0.955	0.949	0.947	0.949	0.979
Wine	0.747	0.747	0.746	0.748	0.746	0.747	0.767
Digits38	0.933	0.932	0.936	0.933	0.937	0.937	0.978
Ionosphere	0.885	0.885	0.885	0.885	0.886	0.881	0.934
Pima	0.737	0.737	0.737	0.735	0.733	0.737	0.770

Table 7 ACP-Gini minus CART per data set with 95% corrected-resampled intervals and Holm-adjusted p -values (pp, percentage points).

Data set	Redundancy change (%) [95% CI]	Holm p	Accuracy change (pp) [95% CI]	Rank-corr. change [95% CI]	Holm p
WDBC	-12.1 [-15.1, -9.2]	<0.001	0.06 [-1.66, 1.78]	-0.057 [-0.085, -0.029]	0.003
Diabetes	-3.0 [-3.9, -2.1]	<0.001	0.23 [-3.17, 3.63]	+0.015 [-0.014, +0.044]	0.782
Sonar	-10.8 [-13.0, -8.7]	<0.001	-0.65 [-4.85, 3.55]	-0.040 [-0.053, -0.026]	<0.001
Cultivar	-13.7 [-20.3, -7.1]	0.001	-0.37 [-2.52, 1.79]	-0.025 [-0.070, +0.021]	0.782
Wine	-29.6 [-31.6, -27.7]	<0.001	-0.10 [-1.39, 1.19]	+0.179 [+0.143, +0.214]	<0.001
Digits38	-7.6 [-9.3, -6.0]	<0.001	0.37 [-2.30, 3.05]	-0.078 [-0.097, -0.058]	<0.001
Ionosphere	-12.9 [-16.8, -8.9]	<0.001	0.09 [-2.77, 2.96]	-0.016 [-0.056, +0.024]	0.782
Pima	-0.9 [-1.4, -0.4]	0.001	-0.43 [-1.49, 0.62]	+0.017 [-0.004, +0.037]	0.397

$p < 0.001$ in each), and the share of leaf samples reached through a path containing a pair with $|r| > 0.7$ fell from 86.2% to 75.8% on WDBC, from 18.6% to 8.3% on Sonar, from 7.9% to 1.8% on Diabetes and from 11.7% to 0.02% on Wine. Pima, the control with no strongly correlated pairs, changed by 0.9%, which is statistically detectable but practically negligible. With α tuned inside each fold, the reductions were of similar or larger size (mean R_w 0.268 against 0.284 for the consensus value, Table 5) and all eight remained significant after Holm adjustment, although the WDBC result was marginal (Holm $p = 0.039$).

Because the penalty uses the same Pearson matrix as R_w , a reduction in R_w alone could be largely mechanical. We therefore recomputed R_w on the single fitted trees with Spearman $|\rho|$ and with normalized mutual information (Table 12). ACP-Gini lowered both in all eight data sets (mean R_w 0.341 to 0.313 with Spearman and 0.130 to 0.123 with mutual information). The mutual-information reductions were small, 0.9% to 2.8% on six data sets, and large only on WDBC (16.0%) and Wine (26.8%), so most of the effect on nonlinear dependence comes from those two. We did not run a user study of audit quality, so the practical value of these reductions for human reviewers remains untested.

Figure 4 shows where the removed correlation mass sits. In CART trees the path feature pairs in the bands $0.5 \leq |r| < 0.7$ and $|r| \geq 0.7$ were 3.8% and 1.2% of pair

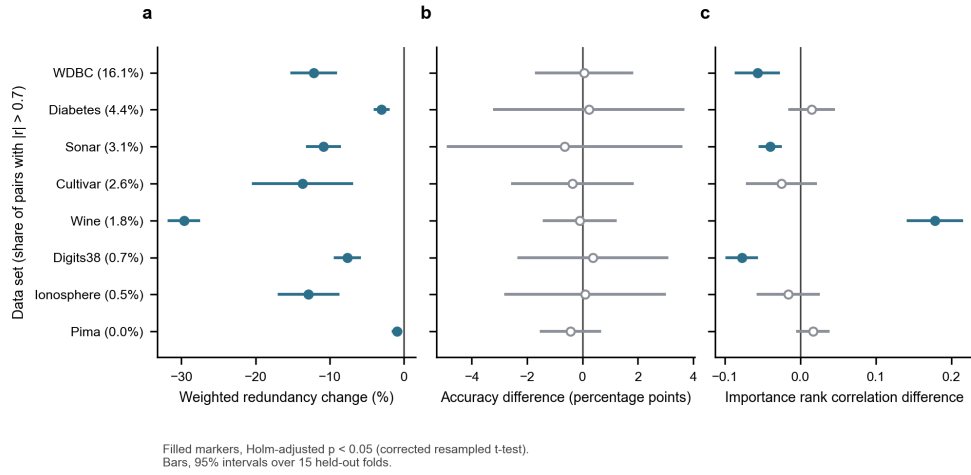


Fig. 3 ACP-Gini minus CART per data set, ordered by the share of feature pairs with $|r| > 0.7$: (a) relative change in bootstrap importance-weighted redundancy, (b) accuracy change, (c) change in importance rank correlation. Intervals are 95% corrected-resampled- t intervals over 15 folds; filled markers have Holm-adjusted $p < 0.05$.

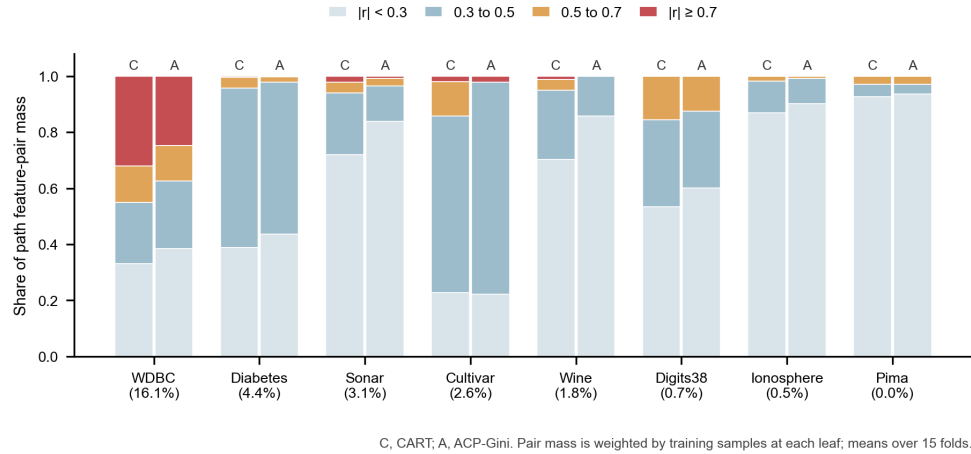


Fig. 4 Distribution of feature-pair mass on root-to-leaf paths over correlation bands, CART (C) against ACP-Gini (A) at the selected α (single trees fitted on each training fold), ordered as in Fig. 3.

mass on Wine; under ACP-Gini both were empty, and the mass moved to $|r| < 0.3$. On WDBC the share of pair mass with $|r| \geq 0.7$ fell from 32.1% to 24.7%. Thus the large Wine reduction is not an artefact of the importance weights; it reflects the elimination of moderately correlated pairs from shared paths. In the 15 CART trees three pairs made up the $|r| \geq 0.5$ mass, density with alcohol ($r = 0.69$, 53%), free with total sulfur dioxide (0.72, 25%) and residual sugar with density (0.55, 23%), and no such pair remained in any of the 15 ACP-Gini trees (pair-level output in `results/path_pairs_high_corr.csv`). It is also the data set whose selected α is the grid maximum (1.0). In the single-tree sweep the Wine reduction was 5.8% at $\alpha = 0.2$ and 29.1% at $\alpha = 1.0$ (Fig. 5), so most of that effect comes from the strongest penalty.

3.4 Comparison with baselines

Table 8 compares importance-weighted redundancy across methods and Table 9 tests ACP-Gini against each baseline. ACP-Gini had the lowest point estimate on seven of

Table 8 Bootstrap importance-weighted redundancy R_w by method (lower is better; best per data set in bold).

Data set	CART	VIF+CART	Cluster+CART	RRF-style	ACP-Gini
WDBC	0.621	0.343	0.526	0.603	0.545
Diabetes	0.352	0.344	0.357	0.353	0.341
Sonar	0.218	0.195	0.207	0.216	0.195
Cultivar	0.453	0.450	0.452	0.451	0.391
Wine	0.183	0.169	0.167	0.181	0.128
Digits38	0.375	0.366	0.369	0.378	0.346
Ionosphere	0.168	0.167	0.166	0.166	0.146
Pima	0.180	0.180	0.180	0.180	0.178

Table 9 ACP-Gini minus each baseline in bootstrap R_w , with 95% corrected-resampled intervals and Holm-adjusted p in parentheses (negative favours ACP-Gini).

Data set	vs VIF+CART	vs Cluster+CART	vs RRF-style
WDBC	+0.202 [+0.141, +0.263] (<0.001)	+0.019 [-0.014, +0.053] (0.230)	-0.058 [-0.076, -0.040] (<0.001)
Diabetes	-0.003 [-0.007, +0.001] (0.240)	-0.016 [-0.021, -0.011] (<0.001)	-0.012 [-0.014, -0.010] (<0.001)
Sonar	-0.000 [-0.008, +0.008] (0.961)	-0.012 [-0.020, -0.005] (0.006)	-0.021 [-0.028, -0.015] (<0.001)
Cultivar	-0.059 [-0.092, -0.025] (0.006)	-0.061 [-0.086, -0.035] (<0.001)	-0.061 [-0.091, -0.030] (0.002)
Wine	-0.041 [-0.044, -0.038] (<0.001)	-0.038 [-0.043, -0.033] (<0.001)	-0.053 [-0.057, -0.049] (<0.001)
Digits38	-0.020 [-0.025, -0.015] (<0.001)	-0.023 [-0.032, -0.014] (<0.001)	-0.032 [-0.039, -0.025] (<0.001)
Ionosphere	-0.021 [-0.027, -0.015] (<0.001)	-0.019 [-0.026, -0.012] (<0.001)	-0.020 [-0.027, -0.014] (<0.001)
Pima	-0.002 [-0.003, -0.001] (0.005)	-0.002 [-0.003, -0.001] (0.004)	-0.001 [-0.002, -0.001] (0.005)

eight data sets, but the tests temper that tally. Against VIF+CART it was significantly lower on five data sets, indistinguishable on Sonar and Diabetes, and significantly higher on WDBC. Against Cluster+CART it was significantly lower on seven and indistinguishable on WDBC. Against RRF-style it was significantly lower on all eight. On WDBC, the only strongly collinear data set, VIF+CART reached 0.343 against 0.545 for ACP-Gini, by dropping 13 of the 30 features on average (17 of 60 on Sonar); ACP-Gini keeps every feature available to every branch. Averaged over data sets, VIF+CART (0.277) was slightly below the consensus ACP-Gini (0.284) and above the nested ACP-Gini (0.268). RRF-style and CART were nearly identical on real data (0.316 against 0.319), so the high synthetic concentration of RRF-style does not transfer. Mean accuracy was within 0.3 points of CART for every method (Table 5).

The filters were run at fixed conventional thresholds while α was tuned, so Section 3.7 sweeps their thresholds on the same folds. VIF+CART was also the most stable method on average (rank correlation 0.488 against 0.454 for CART and 0.453 for ACP-Gini) and was above ACP-Gini on Diabetes, Sonar and Digits38 ($p < 0.01$ each), whereas ACP-Gini was higher only on Wine; this metric counts dropped features as unused, which favours a method that removes features.

3.5 Stability

Stability did not improve uniformly. Importance rank correlation was significantly lower than CART’s on WDBC (−0.057), Sonar (−0.040) and Digits38 (−0.078), significantly higher on Wine (+0.179), and not significantly different on the other four (Fig. 3c, Table 7). Across data sets the top-five and rank-correlation metrics were lower in 5 of 8 and the used-feature Jaccard in 8 of 8 (Table 10). The Wine gain is therefore metric-dependent, since its used-feature Jaccard fell from 0.986 to 0.898 while its rank correlation rose.

Table 10 Number of data sets (of eight) in which ACP-Gini is lower than CART, with a two-sided sign test.

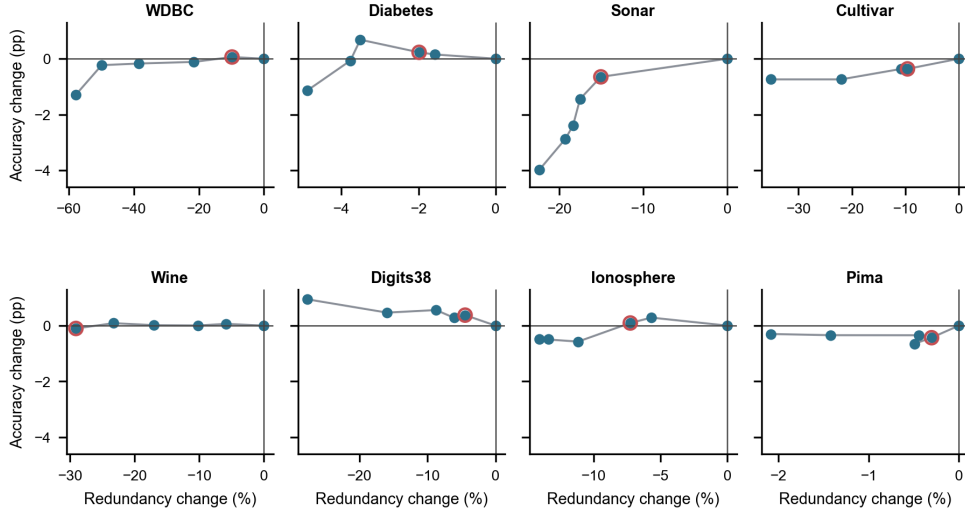
Metric (ACP-Gini lower than CART)	Data sets	Sign-test p
Weighted redundancy	8/8	0.008
Mean path $ r $	8/8	0.008
Paths with a pair $ r > 0.7$	6/8	0.289
Importance rank correlation	5/8	0.727
Used-feature Jaccard	8/8	0.008
Top-five Jaccard	5/8	0.727
Group-level importance correlation	6/8	0.289
Group-level Jaccard	8/8	0.008
Accuracy	4/8	1.000

One could read the loss as harmless, a reshuffling among interchangeable members of the same correlated block. If so, the rank correlation of group-aggregated importance would be unchanged. It was not. Groups are formed from the full-data correlations as defined above, and Pima has none with more than one member, so its group metrics repeat its feature-level ones. On the other seven data sets the group-level importance correlation was lower under ACP-Gini on six, significantly so on WDBC (0.677 to 0.520), Sonar (0.235 to 0.173) and Digits38 (Holm $p < 0.001$), and higher on Wine (0.752 to 0.850). The group-level Jaccard was lower on all seven. We therefore report a real stability cost on the data sets with the most correlation structure. A plausible mechanism is that the penalty makes later choices depend on earlier ones, so a bootstrap perturbation of the root propagates through penalized children, but we have not tested this, so it remains a hypothesis.

3.6 Robustness to α , depth and correlation estimator

Operating curves in Fig. 5 (single trees fitted on each training fold) show how far α moves the trade-off. On WDBC, $\alpha = 0.6$ cut importance-weighted redundancy by 38.5% with an accuracy change of -0.2 points, and $\alpha = 1.0$ cut it by 57.8% at -1.3 points. On Sonar the same range cost up to 4.0 points, and Wine and Pima changed accuracy by less than 0.7 points. The consensus value is deliberately conservative. A depth cap of 3, 4, 6, 8 or 10 left the sign of the redundancy reduction unchanged in every data set except Pima at depth 3, and the accuracy change stayed within 1.4 points in every data set and depth cap (Fig. 6).

Table 11 varies correlation type, aggregation and scope at the selected α , again on single fitted trees. Pearson and Spearman correlations gave the same redundancy reduction (8.8% and 8.9%), min and mean aggregation gave slightly less (7.4% and 7.0%), and the node-local scope gave the least (6.4%) at a small accuracy cost. Accuracy differed by at most 0.35 points across all configurations. Product aggregation with global Pearson correlation is therefore the simplest default with the largest effect, and the choice between Pearson and Spearman does not matter here. Differences among configurations were small compared with the effect of α itself.



Each point is one alpha in $\{0.2, \dots, 1.0\}$ relative to $\alpha = 0$ (CART); red ring marks the selected alpha.

Fig. 5 Accuracy against weighted-redundancy change for $\alpha \in \{0.2, \dots, 1.0\}$ relative to CART, mean over 15 folds. Single trees fitted on each training fold. The red ring marks the selected α . The accuracy axis is shared across panels.

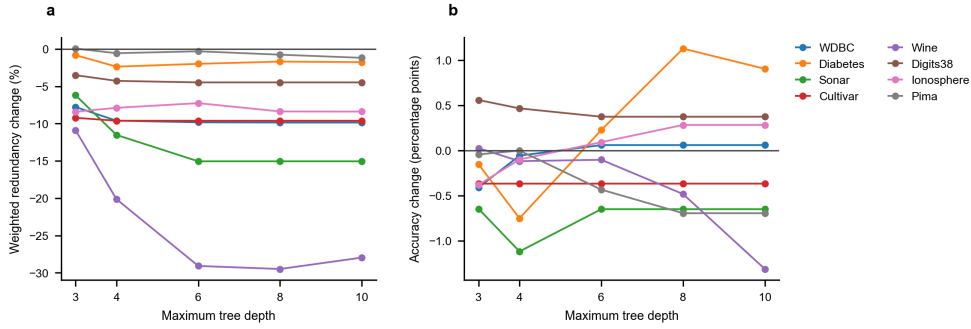


Fig. 6 Sensitivity of ACP-Gini to the depth cap: (a) relative change in importance-weighted redundancy and (b) accuracy change against CART at the same depth cap. Single trees fitted on each training fold.

Table 11 Ablation of correlation type, aggregation and scope on single trees fitted on each training fold, averaged over data sets (15 folds each); R_w is not the bootstrap average of Table 5.

Configuration	Accuracy	Weighted red.	Change (%)	Path $ r $
CART ($\alpha = 0$)	0.8248	0.323	0.0	0.288
Pearson, product, global (default)	0.8238	0.294	-8.8	0.249
Spearman, product, global	0.8234	0.294	-8.9	0.251
Pearson, min, global	0.8250	0.299	-7.4	0.254
Pearson, mean, global	0.8260	0.300	-7.0	0.256
Pearson, product, node-local	0.8225	0.302	-6.4	0.265

3.7 Does path-local scope matter, and how do tuned filters compare?

Two checks address the weakest points of the comparison. First, we replaced the path-local penalty by a correlation-aware penalty over every feature used anywhere so far in the tree. It gave the same average redundancy (0.294 for both with Pearson,

Table 12 Scope ablation and filter-threshold sweeps on single fitted trees, mean over the eight data sets of the per-data-set fold means. R_w uses the Pearson, Spearman or normalized-mutual-information (NMI) matrix; features kept counts the features available to the tree.

Configuration	Acc.	R_w Pearson	R_w Spearman	R_w NMI	Features kept
CART	0.825	0.323	0.341	0.130	27.5
ACP-Gini (path-local)	0.824	0.294	0.313	0.123	27.5
Correlation penalty, tree-global	0.824	0.294	0.313	0.122	27.5
VIF+CART, threshold 5	0.820	0.262	0.283	0.105	20.1
VIF+CART, threshold 10	0.823	0.294	0.311	0.120	23.1
VIF+CART, threshold 20	0.823	0.307	0.327	0.123	25.0
Cluster+CART, $ r \geq 0.5$	0.827	0.302	0.317	0.123	15.3
Cluster+CART, $ r \geq 0.6$	0.824	0.302	0.317	0.123	17.9
Cluster+CART, $ r \geq 0.7$	0.822	0.311	0.330	0.126	20.8
Cluster+CART, $ r \geq 0.8$	0.823	0.315	0.335	0.127	23.1
Cluster+CART, $ r \geq 0.9$	0.824	0.320	0.339	0.128	25.8

0.313 with Spearman, 0.123 against 0.122 with mutual information) and the same average accuracy (0.824), with per-data-set differences in both directions (Table 12). On these data the path-local scope is therefore not shown to be necessary, and the active ingredient is the correlation-aware discount of the gain. Second, we swept the filter thresholds on the same folds. At threshold 10, VIF+CART matched ACP-Gini on average redundancy (0.294) with 23.1 of 27.5 features kept on average, and at threshold 5 it was lower (0.262) with 20.1 kept and 0.5 points less accuracy. Cluster+CART at $|r| \geq 0.5$ reached 0.302 with 15.3 features kept. When tuned or swept, the filters are thus competitive with ACP-Gini on redundancy and differ mainly in discarding features, which ACP-Gini does not. Each sweep point is a different operating point, so we do not rank them.

3.8 Tree size, runtime and a worked case

Tree size changed little. The mean node count of ACP-Gini was within 2.7% of CART’s on every data set (24.2 against 24.7 on WDBC). Fit time at $\alpha = 0.6$ was 0.90 to 1.35 times CART for the global scope and 1.01 to 1.42 times for the node-local scope, with mean fit times under 0.08 s on one machine; ratios below one reflect timing noise, and the overhead is small. On WDBC, CART and ACP-Gini disagreed on 53 of 1707 test predictions across the 15 folds (3.1%). ACP-Gini was right on 27 of these and CART on 26, and 16 lay near the $[0.4, 0.6]$ probability boundary, so the tree change does not systematically favour either model.

Table 13 traces the mechanism at the right child of the WDBC root, whose path ancestor is worst concave points (Fig. 7). CART picks mean texture on raw gain. Its correlation of 0.353 with the ancestor gives a penalty factor of 0.929. Texture error has slightly lower raw gain but a correlation of 0.112 with the ancestor, so its factor is 0.978, and ACP-Gini selects it. Mean texture is moderately, not strongly, correlated with the ancestor, which is the regime where the penalty reorders near-ties without suppressing informative splits.

4 Discussion

ACP-Gini lowers redundancy consistently, in the selected feature set and along decision paths, without a significant accuracy change on any of eight data sets, and the reduction persisted under Spearman and mutual-information measures. The size of the benefit follows the correlation structure of the data and the penalty strength rather

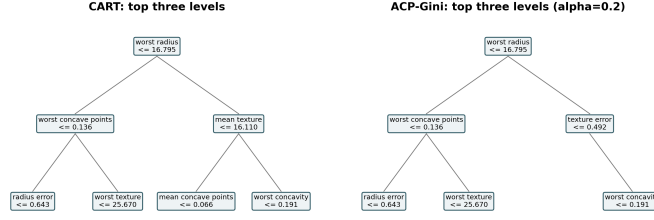


Fig. 7 Top three levels of full-data WDBC trees at the selected $\alpha = 0.2$.

Table 13 Candidate split evaluations at the WDBC root’s right child (selected $\alpha = 0.2$).

Feature	Raw Gain	Corr with Root	Penalty	ACP Score	CART Rank	ACP Rank
mean texture	0.0437	0.3526	0.9295	0.0406	1	2
worst texture	0.0437	0.3599	0.9280	0.0405	2	3
texture error	0.0421	0.1117	0.9777	0.0412	3	1
mean concavity	0.0412	0.6882	0.8624	0.0355	4	4
worst concavity	0.0372	0.5740	0.8852	0.0330	5	5

than a fixed rate. It is largest where moderate correlations cluster on shared paths (Wine) or where collinearity is high (WDBC), and negligible on the Pima control. The 29.6% on Wine is a bootstrap figure at the grid maximum $\alpha = 1$; the single-tree sweep gives 5.8% at $\alpha = 0.2$.

Two findings narrow the claim we can make for the method. A correlation-aware penalty over the whole tree gave the same average effect as the path-local penalty, so the evidence supports the discount itself and not the path-local scope. And when the filter baselines are swept or tuned they are competitive on redundancy: VIF pruning at threshold 10 equals ACP-Gini on average, lower thresholds do better, and on WDBC both filters beat the consensus ACP-Gini. What ACP-Gini offers over a filter is that no feature is removed, so a variable can still be used on a branch where it carries conditional information. Whether that availability improves a model is a question for data with such conditional structure, which we did not construct.

The stability result changes how the method should be presented. The penalty improved importance-rank stability only on Wine and lowered it on three data sets, and it lowered used-feature stability on all eight. The group-level analysis shows that the loss is not merely swapping within blocks. For auditing, where the decision rules of one fitted tree are reviewed, this may be acceptable; where the identity of the top features must be reproducible across resamples it is not. The operating curves let practitioners choose α against both costs.

4.1 Limitations

The evidence has limits that the reader should weigh. All data sets are small, tabular and binary, only WDBC is strongly collinear by the $|r| > 0.7$ criterion, and none is a regression or multiclass task. Every statement about the strongly collinear regime, including the VIF advantage and the largest stability loss, rests on that one data set and should be read as anecdotal, and we cannot claim effects on high-dimensional or heavily collinear domains. Three of the eight data sets were added after the original five and no data-set selection was pre-specified. The accuracy intervals are wide on the smallest data sets, so losses of several points are not excluded, and the random-forest

gap of about four points is specific to its depth cap of 6. The redundancy measures are computed from the same correlation matrix the penalty uses, which makes them aligned with the objective rather than independent measures of interpretability; we did not run a user study of audit quality. The main comparison does not match tuning, because α is tuned while the VIF and cluster thresholds are fixed at conventional values; the threshold sweeps in Section 3.7 show that this understates the filters, but they report operating points rather than a tuned selection protocol. The 15 folds overlap, and the corrected test, derived for test error, is applied here to bootstrap-derived metrics as an approximation whose calibration we did not verify. The Holm adjustment covers the eight data sets within each metric and baseline, with no family-wise statement across the many metrics we report. ACP-Gini is a greedy heuristic without a stability guarantee, linear correlation cannot separate redundancy from useful nonlinear interaction, and a very strong penalty ($\alpha = 1$) can expose weak splits on noise features.

4.2 Implications and future work

For auditable single trees, ACP-Gini offers a path-local way to avoid presenting the same evidence under several feature names, with a tunable price in stability and, on small data, possibly in accuracy. Future work should test non-linear association measures such as mutual information or distance correlation, categorical associations with Cramér’s V, regression trees, and use in forest ensembles. A controlled study of reviewers auditing trees, a stability objective inside the criterion, and evaluation on high-dimensional genomic or tabular benchmarks would resolve the main open questions above.

5 Conclusion

ACP-Gini reduces correlation redundancy in single decision trees, along root-to-leaf paths and in the selected feature set, reduces to CART at $\alpha = 0$, and showed no detectable accuracy change on eight data sets, although small losses are not excluded. It does not improve stability in general, a tree-global correlation penalty gave the same average effect, and tuned feature-removal filters are competitive on redundancy, so the contribution is a correlation-aware split discount that keeps all features available, with a controllable trade-off among redundancy, accuracy and stability, for models that humans must audit.

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